

ActInf GuestStream 083.1 ~ Simon Wardley

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all right hello and welcome everyone this is active inference guest stream 83.1 here with Susan Hasty and our guest Simon wordley so Susan will say a few words that will lead us in Simon has many resources prepared and exciting perspectives to share so thank you both for joining Susan to you okay are you ready for me go for it awesome okay thank you so much Simon I was been looking so forward to uh to introducing you to the active inference Community um and uh in Wordly mapping uh and what you have been able to um accomplish and you know it is is amazing how much admiration um that uh that people have shown you there and so I'm gonna just I'm gonna I'm British you can't do this to me you can't do this to me we're not American so is that you know that actually there's a connection between being the most strategic Brilliant Minds that people know and the humility you bring to it along with the uh the jovial it's a cultural thing so then we are so I turn it over turn it over and and but we would like to I would like to you to share how you came about um you know the the um moving from the you the experts on the stage to how do you how did you tap into the need for Bly mapping what was that where did that where did that arise from well well it was obviously some Fine Day in the UK and I was sitting down there with my enormous mind uh thinking about these problems and and naturally the pure Genius of myself I I no no it was complete random accidents and everything else I had got a clue what I was doing so I um I was running a compy at the time um and uh well actually I've got a I've got some slides and give you a bit of the the sort of formal history I can show you the horror of it if you wish um can you Daniel is it possible for me to show my slides thank you very much uh so I'm not going to go through this whole it's 127 slide deck I'm not going to do that I mean um uh but what I will do is just talk about a couple so I'm going to talk about the origin of mapping and then we'll get on to I may as well explain what a map is what it isn't and then I'll mention a little bit about patterns and then we've got these other topics change gameplay how you identify value but rather we won't go down that path we'll we'll get into a a different style of conversation I can show you something called Meo and um because Susan you are part of my education Group which I really really appreciated way um helping us map out that space so let's start with the origin

since you asked me that question um this is company called Fango uh that I used to work for I got a photo this of the website go see the CEO um and it was a very uh sort of successful company at the time uh very very profitable uh rapidly growing uh but it had a problem and that that problem was the CEO uh the fat cat in charge had absolutely no idea uh what they were doing they were just making it up uh I mean um they were totally and utterly clueless and I know this for a fact because I was the CEO I mean I I used to um come up with all these wonderful statements about what our future was uh so this is 2004 uh our strategy as customer focus we will lead an Innovative effort in the market through our use of agile techniques and open source uh become a leading provid oh horrible but I literally pinched these words uh from another company and just changed a few things um and I I I sort of was you know I had no idea what I was doing I used to go around listening to other CEOs talking about strategy and I used to record all the words they would say and I I started aggregating these words and I called these uh aggregations business level abstractions of a healthy strategy or or blahs for short and I used to publish these common blah lists where things like you know these are the words that people are using digital disruptive Innovative collaborative competitive Advantage ecosystem open source Network blah blah blah and so then what I did was I created What's called the blah template straty is blah we will lead a blah effort of the market through our use of blah and blah to build a Blah by simply grab all these sorts of vision statements and smashing them together until I came up with a generalized form and then I would just combine the Blas and the blah templates uh to to autogenerate strategies so uh you know um last time I did this I created 64 things like this our strategy is customer focus we will lead a destructive effort of the market through our use of innovative platforms and data to build a collaborative ecosystem it sounds it's complete garbage um but I used to send these around to people and I I get I get three basic responses back uh which are the form Well here here's another example our strategy is innovative digital business and we load a growth effort of the market through our use of customer and I can barely read it that is that painful uh the three basic forms I used to get back was this is the exact wording from our business plan I've seen two of these used already and my favorite one was aru for five which you know I didn't want to tell the people but I just autogenerated this garbage um so I was no idea at all what I was doing completely clueless I had done an MBA and of course I assumed back in those days you learned all the secrets of how to understand what a business was is uh from doing an MBA I now teaching um couple of days here and there every year at different Business Schools and I realized the delusion that I had over what you actually learned um so you know I knew I was clueless I

started to think maybe other people were struggling um a friend of mine by the way if you ever want to I I don't think it's available anymore this is a little bit of an old presentation um he actually cre just automated this there was a uh lap that was built that would would create you one of those generic sort of it's much easier these days you just go to chat GPT and just say give me a strategy and it'll give you some garbage and and you're away um so I ended up reading every book I could find I was getting nowhere and by pure luck I I I was in a a booksh shop in Ching cross in London and I was talking to the Book seller and I I was talking about all these straty books I'd read and how none of them made sense and she asked me had I ever read um The Art of War by Sun Sue and I hadn't and so she persuaded me being a fantastic book seller to buy two different versions of the book because they're all translations got to say I am so grateful uh for that because it was in the reading of the second version I noticed a particular pan so sunu talked about five factors that mattered in competition now when we talk about competition competition is the act of a group of people seeking something um whether it's knowledge resource or whatever it happens to be and there are many different forms of competition so sometimes we will conflict with each other that's we fight each other over something sometimes we will cooperate with each other so helping each other sometimes we will uh collaborate so laboring uh with each other so sometimes fighting working or or or laboring so those are multiple different forms of competition uh they're all competition they're all part the act and normally uh we will possibly collaborate with one group over here in order to cooperate with another group over there in order to conflict with another group over there so you can have many many different forms combinations of this I often hear people talk try and create new words for this I mean um um mostly because they want competition just to mean conflict and therefore they can have different words like collaborate which which is no no longer competition because they Define competition as only conflict and then cooperate and so then they get to things well if we collaborating and conflict but we don't call it conflict we call it competition we're co- competing collaborating and then you get into these large horrible spaghetti words just meaningless it's really simple though competition it's the act of seeking in multiple groups and there multiple different forms so when you look to the um Sun su's work talk about five factors that matter one have a purpose a moral imperative to what you're trying to do secondly understand the landscape that you're competing in uh uh and then third understand the climatic pain so the heavens the weather so how the landscape is changing and then you get into uh principles of how you organize yourself Doctrine and then you get into the sort of gameplay and Leadership and I I was like fascinated by this and then I

came across John Boyd the Mad major uh famous for creating the UDA Loop and so the first o of the UDA Loop is to observe the environment so that's what landscape and climactic patterns are about or observation of a space then you orientate yourself around the space that's what principles are about and then you need to decide where you're going to act and this is where we get into the game play and all those sorts of leadership choices and at the heart of this a fundamentally two WI the why of purpose and the why of movement why do I make this decision over over that decision so this became for me the the the you know the strategy cycle because it sort of made sense uh you know I remember I hadn't done the MBA so so I didn't know the official things to talk about and I got really fascinated by this idea of observing the landscape and so that's what brought me into to Maps because I used to read everything I could find on sort of military history and one of my favorite battles is the uh Battle of themop so the ancient politicians well um theistically ancient politician Greek General had a problem um they had the the Persians were invading and we don't know quite the number of Persians they're about 100 well we think somewhere between 100,000 to 70,000 roughly um and so the Greeks were independent city states and the mystically was I believe Athenian and so what they decided to do was you know how are we going to cope uh with this Invasion force uh what they decided to do was to block off the streets of Artesia Force the Persians along a coastal Road into an nrow pass called theropo uh where you know a small number of TR could defend against a larger force and in fact the number of troops there there were about 4,000 Greeks there and they were independent city states but they they came together at times um and within those 4,000 with 300 SPS and this is where we got the story of the 300 from uh the 300 defending for a considerable amount of time against a much larger Force now my fascination with this m was was um I could use it as you know we could use it as a way of discussing what we're going to do what do we need to move where do we need to defend uh who do we need to work with it's a vehicle communication and I started looking at my own business and thinking about how do we communicate and make decisions and back in those days I was using something called a swap diagram strengths weaknesses opportunities threats be oh fantastic you know um so I created a SWAT diagram for this battle so strength a well-trained Spartan Army a high level of motivation not to become a Persian slave weaknesses the e4s might stop the Spartans turning up a truckload of Persians are turning up opportunities get rid of the Persians um and also get rid of the Spartans where we happen to be athenium we don't actually like the Spartans as I say independent city states uh and the threats and the Persians get rid of us and the Oracle says a really dodgy film might be produced a few

thousand years later obviously added that one a bit later um but I simply put these against each other and I ask myself you know what would you communicate and you know use to communicate and determine strategy in battle and you know the same goes for what would you do in saying competition um you could either use position and movement through an understanding of the landscape that you're competing in or you can use some sort of magic framework like a swap just thinking well crank it would be obvious you would try to understand the landscape before you you make a decision and here I was making all these decisions using a magic framework so at that point I said right I need to understand the landscape that my business is operating I need a map I went looking for maps of uh business and I couldn't find any um well interestingly I can find well I found lots of things called Maps so I I I asked for example in my company everybody to send me any Maps they had and they said would send me loads like things like systems Maps or business process maps and mind maps and lots of stuff like that but when I took one of these maps and this is a map of uh simple map of the uh online photo service and I I looked at one component CRM which stands for customer relationship management and I simply moved it across the screen but kept the links the same so I went from there to there and asked how has the map changed but the answer is it hasn't changed and for me that was immediately on OD because I know if I take a geographical map of the world and I take Australia and put it next to the UK that has fundamentally changed the meaning of the map so why is the meaning of the map not changed here well it's because it's not a map it's a graph and that's what all these maps that I had had in common none of them were Maps they were all graphs uh business process graphs mind graphs they're all nodes and connections between nodes rather than maps of the landscape so just to really spell that out um these are uh three graphs uh and they are identical there are three places in the UK Nottingham London DOA connected by two roads M1 M2 now the thing about a graph all that really matters in a graph is the nodes and the connections and the waiting on the notes that's it um the difference between that and a map is the reason why the three three maps at the bottom are all completely different while the three graphs at the top are identical is the in a map space has meaning you can't actually just move a component without changing the fundamental meaning of the map itself so that's the distinction between a graph and a map and everything I had which was called a map was in fact a graph so I I I sort of assumed at that point well this must be what you learned at MBA schools um because obviously you learn how to do the mapping and there are no books on the subject uh so it must be like really secretive and really useful um or uh or or maybe completely useless and have'

been abandoned long ago you I just sort of didn't know but nonetheless I had no access to those those resources so um and I couldn't find any examples so I I thought well I've somehow got to create a map so I looked at a map and there basically three components you need you need an in order to give landscape the space meaning you've got to have an anchor such as magnetic north you've got to have position of pieces so this is North Southeast or west of that and then you've got to have something known as consistency of movement so if I'm going north I'm going north I'm not going south no sort of teleportation there's got to be some sort of consistency so um I thought right let's try and map something and so I did and I MA my business but I I'll show you a t-shop only because my business bit was bit dull and t- shops I find more fascinating don't know why so what's my Anor going to be so I thought well I'm going to focus around the users and immediately well who are the users I've got my consumer and I I've got the business I've also got regulator I've got many users okay well first we better identify who the users are so we start so we'll start off with the public and the business okay well they have needs and hopefully the public has a need for a cup of tea if I'm doing a tea shop and the business has a need to sell Cups of Tea because that's where it's going to get its money from all right uh so my Anor are going to be public in business um I can put Regulators here okay uh so what what next and I was thinking about this when I thought a cup of tea has needs itself it needs a cup it needs tea and it needs hot water and of course hot water needs cold water needs Kettle and Kettle needs power so what I've got is a chain of needs this needs that and the further I go down that chain the less visible something becomes so if I'm a member of the public drinking my cup of tea the power that went into the sort of kettle that's quite far removed from me so I've got this notion of position through how visible something is how and it's described itself in the chain okay but this is a graph I thought how do I turn this into a map well it turns out cuz I was looking at this stuff and think thinking about well cattle and power um I realized you know power hasn't been uniform it's changed over time well like the kettle you know the electric kettle for example obviously somebody in it's now a commodity but at some point somebody invented it so all of these components turned out to be different forms of capital and they all seem to be evolving and so I did a piece of work looking into how things evolve and I stumbled um in the British library after six months of working on this problem I I had a whole bunch of data and it came together and it just fell in place a lot of accidents and it turned out there was a fairly simple process by which things evolve you start off with the Genesis of Novel and new items then you get custom built examples of things then you get products and rental services and then you get commodity and Utility Services

now these are just labels for different stages 1 2 3 and four because it turns out you can map things like data knowledge ethical values all sorts of things but what I've got now on this map is I've got the anchor at the top business and public I've got position described through a chain of needs and I've got this concept of movement or Evolution um in terms of how evolve the forms of uh um capital r so this was the sort of first map that I produced back in well actually it was uh because I'd actually mapped my own business so I was just finding a way of explaining somebody else had to do mapping so wasn't quite the first but one of the first Maps uh that I produced in 2005 and I was looking at this and I realized um what I'm actually doing is I'm exposing my assumptions about how to make a t- shop in a way that other people can look at those assumptions and Challenge and question what I'm doing because you might go why is kettling custom Bill surely kettle's more of a commodity you might go we're a staff uh you might go hey we're a business we got to take payment we're missing tills we're missing somewhere for people to see uh you know we to sit sorry we're missing um you know what about cakes um there's all sorts of things um what I'm doing is I'm exposing my assumptions in a way that people can see the sort of knowledge I have around a space and they can add to it and something bizarre I realized is that in most the organizations I'd ever worked in uh we were all very much uh story Le organizations and in fact there's an entire industry which tends to go around telling people that to be a good leader you've got to be a good Storyteller well the problem with that is if I give you a story and if you challenge my story you're challenging my ability to leave so I tend to get very defensive but if I put my story in a map you're not challenging me you're telling me there's something wrong with the map so suddenly I found a loss of the sort of tensions which we were to have around discussing something just just disappeared because we we could discuss the problem uh through the medium of the map and I've done this with things like uh brexit and remain in the UK to all sorts of really thorny issues getting quite difficult groups to uh find ways of actually uh talking to each other through the medium of a map and it's it's it's not it's an old architecture trick it's why use models uh models of buildings as a as a way of getting people to to to look at and describe the you know their feelings about things and their assumptions about things um so I started using these maps and I used to go into all sorts of uh organizations and uh I I was working with these maps and I ended up doing all sorts of interesting things in government as well so this is a friend of mine Liam Maxwell we wrote something called the with several others the better for Less paper Francis M which led to a whole bunch of transformations in UK government um and just simply by mapping out one particular project this was back in about 2014 I mean

they they saved about 425 million um and they estimated about 1.5 billion in its lifetime now this these might s well some ridiculously huge fig how can this possibly be true I mean my maps have been used to put up uh NASA and Planet work together through through my full mapping used it to put up the uh interestingly carbon mapper satellite and just to show you why this sort of stuff uh these impact big impacts become possible uh this is an insurance company this was back in roughly 2011 200 12ish um this is a business process flow um for putting servers into their data center so they would they needed compute they'd order a server server would go into Goods in modify Mount Racket and they had a bottleneck in this process flow and so they' invested time about six months um got various uh uh vendors in and worked out a way using robotics to to get rid of this bot onic so it would accelerate the speed at which they could you know rack service basically and um it's quite impressive work return of investment calculations sub one year uh several million of capital expenditure in terms of the robotics they're just about to go for it and I was asked to come and have a look at this problem um it was a friend of mine just asked me to just go and have a have a look just make sure it Mak sense and so the problem is they' spent six months working on the problem um they got all their vendors they got all their business case they got all their calculations I mean I'm just turning up I'm nobody um if I challenge the story you know it's a game over so I said well I don't really know what you're doing do you mind if we just map it and they were like oh we don't don't really want to m it and blah and I think we spent about 20 minutes arguing about whether we should map it and then about 15 minutes actually mapping it and so this is the map they came up with there we are 2010 slightly ear than I thought user needs compute which they yeah put in product order server very commodity uh servers Bly enough they thought servers were commodity but compute was slightly less okay it's 2010 um it's surprising only because of cloud was 2006 uh Goods in very commodity and then they had rat Mount modify and I was able to look at this map and simply go why have you got rack and custom build and they said oh we have custom build racks ah so what are the modifications you're making to service well the servers don't fit into our custom build racks so we have to take cases off Drew new Hol ad new plates that's the bottleneck and that's why they need robot and somebody in the room simply just went why don't we use stand at recks um now what you've got to remember course when somebody said that everybody's looking at them yeah why don't we uh why are we doing all this customization it just doesn't make any sense at all now what you have to remember these people aren't DED these people are simply trapped by a story um so at some point in the past it made sense uh to have custom built racks because there was no such

thing as standard racks but they never evolve from that point so obviously what you want to do is evolve that component what they'd actually been doing all of this time is improving the process flow so you often see these people focusing on process flow rather than evolutionary flow so you know they literally do the wrong things I I mean I had one organization well actually I've got another one actually this another particular organization they were they had weighing scales so we this particular group had so many paper forms um that it was actually cheaper to weigh the number of paper forms they used to come in palet to calculate the number of paper forms there were than to manually count them and and they need to do this to quickly report a specific number to another group um so they literally had weighing scales for this and they had some mathematicians who worked on algorithms to determine you know how many uh how much weight went to how many paper fought and so simply mapped it out at the top and of course then you look at where the paper forms coming from Goods in uh you go and talk to people in Goods in and the paper forms are coming from distribution sites all over the country enough and they're in this sort of digital program they want to basically make the weighing scales connect to their computer systems so uh uh that's their digital transformation but you go down to a distribution site and ask them what are they doing and they they've got big printing presses because they're printing out paper forms and you go well where are you printing them out from So You map that out and it turns out that users are filling in an online website to which they print out these forms and then they ship them to this other group and they don't know why they ship them but they ship them to this other group who then weighs the forms to work at how many users filled in an online website so the entire process because no one understands the whole picture makes no sense whatsoever so this sort of stuff is typical people don't think about evolutionary flow and one of the things once you start doing this um you start discovering uh you start using Maps before we do something to do what we call premortem challenge so before we start act like investing in robotics we'll map out the space and then we'll ask ourselves you know does it make sense we Challenge and then we'll go and do something and then once we've done something we do postmortem learning using exactly the same vehicle the B and from that we learn Pats now there are so many patterns uh um which there are what we call climactic patterns there's about 30 of these principles we found about 40 about 100 different forms of gameplay most people are oblivious to all of this stuff and I've used it um myself um so I used to run straty for a company called Canon cor they provide something called a buntu uh in 2008 we were 2 to 3% of the operating system market up against Microsoft and red had they had all the money or everything all the

people um we had half a million pounds and we had 18 months and we went from 3% of the operating system Market to 70% of all cloud computing so um if you understand the landscape it can have a devastating effect against those you compete with but this is uh James Finley good friend he used to be the CIO for hs2 highspeed rail big heavy engineering project and James had a problem I think this was about 2012 um that he um well they wanted to build a virtual world of the Railway um because it's cheaper to dig up a virtual world and go whoops we've got that wrong than it is the English Countryside So This Is A System's graph for the system building a virtual world of the Railway and James's problem was he want to know how to manage this so typically in government in those days we used to break it into small contracts uh like lot one engineering lot two user experience we group common things into those contracts and then would Outsource it to some supplier and the net result was almost always the same massive cost overruns massive delays and this sort of stuff so so James wanted to find a different way now I'd been mapping from 2005 and some of the patterns I'd learned directly applied to this model so um I was very fortunate in that James had spent Sunday afternoon um creating a map and I said well send me what your map what you've got so this was the map and this was in 2012 and I said well it's quite easy now to work out how to manage this sort of stuff the reason why it's quite easy is that as things evolve their characteristics change so on the left hand side it's very Uncharted chaotic uncertain and on the right hand side it's very industrialized doesn't matter whether it's money penis and Computing it all evolves from that sort of genst or more commodity and because of that we'd learned back in 2006 uh there's no such thing as one size fits all management methods so things like agile extreme programming were very good um on the left hand side where your focus had to be about reducing the cost of change because change was normal uh where on the right hand side of the map you want to be all about reducing deviation so things like Six Sigma outsourcing good and in the middle of the map you want to be about leaning off the shelf products because you're all about learning and reducing waste so we simply took that and you apply it to the map and you go right we're going to build in house with agile techniques over here we're going to use off-the-shelf products in the middle and we're going to Outsource to utility providers on the right hand side and if we want to we can now break it into small contracts where we bring together things which are similar in terms of nature of how we manage them rather than just random grouping of words because they sound similar so that's what James did and that project delivered way ahead of schedule way under budget ended up in front of the Public Accounts committee um quite an exceptional thing but one of the patterns we learned

well in this case a principle was use appropriate methods it didn't matter whether we were doing financing or or project management there's no such thing as a one-size fits all method because the characteristics change and therefore uh the way you treat it has to change as well of course there's a word of difference between using agile like extrem program and being agile being agile means basically using appropriate methods I mean if you try and use Six Sigma across the lot you'll be good on the right hand side you'll suffer on the left if you try to use agile across the lot you're good on the left and suffer on the right so if we go back to uh the original contract structure and I take one of those contracts lot one engineering and I simply take that and look at the mapping format so exactly the same contract but now I'm looking at the map I can tell you this contract is going to fail before we even sign the paperwork and it's going to fail because the stuff on the right hand side might be efficiently treated we can Define it in a contract but the stuff on the left hand side would always incur excessive change control costs we've doomed out elves by trying to Define that's which we're still learning about so I mean that's a gentle introduction I'm not going to go through all the patterns and everything else but even at this stage I mean the most common problems I find with business um is one they don't actually understand their users which of course you have to do in order to map that's the anger secondly they don't understand their user needs um which is fairly terrifying thirdly they don't understand their comp components are the supply chain of all the things which go into making up what they're actually doing and I think that's become extremely clear over the last 10 years that so few companies actually understand that supply chain um but worse on that you can't just treat every component of the same so they don't even understand how evolved those components are which is why we always end up with these fads of one size fits all let's agile everything let's six sign or everything or whatever it happens to be none of them work because you have to use appropriate methods cuz the components evolve so their characteristics change so along with not understanding uh users user needs not understanding the supply chain components not understanding how evolved components are uh using inappropriate methods um it turns out most organizations that their structures are fundamentally flawed um because then that needs to change with the context and they also don't have any mechanism to effectively challenge each other without Landing in that Dreadful world of stories which is highly political and so you you learn the lesson pretty quickly you don't challenge the executive story well we want to but there is no mechanism to do so without causing that confrontation um so that i' learned those lessons probably about 2007 and it's just grown grown since then and as I say there's about 30 common economic

patterns um which are useful for anticipation it's about 40 principles like use appropriate methods it's just one um and then you start to learn you can manipulate the space and there's about a 100 different forms of gameplay so I suppose that's my as easy as I can make it introduction into mapping at that point I'll stop I won't stop sharing because Daniel can do that for me there we are Susan was that a good enough intro absolutely absolutely actually you uh yeah despite I was actually in three of your projects what education um and and so you despite actually being in the midst of it um what you've showed me today is is that it's you when we talk about components um that may need some clarification uh to be able to translate that into active inference but but the the this is also showing the internal um representation the resource uh utilization which is very important in active inference we refer to that as you free energy um and um and so it's not just about you know what is here what I'm trying to you know either avoid a threat or go after an opportunity exploration exploitation but it's also about how I'm I'm actually organizing or how to look at the organization the internal composition or configuration in terms of that utilization so utilization of opportunity and how do you how do you actually think about that so so help me bring that home for active inference Daniel so so there's a couple of points I can run off that um and because I I I love the Practical examples actually um because that's where I sort of come from I um I mean we can talk about organizational structures I I I developed a cellbase structure which used to be called Pioneer settlers Town planners those words are very problematic so these days call explorers Villages and town planners it it's a uh basically a structure you break using a map you break into small uh groups very much explorers on the left and side it turns out that people have different attitudes and work well in different areas how they have um um also different aptitudes so you get Explorer Engineers explorers marketers explorers in terms of finance and it's good to bundle those people together um and uh what we do is then we introduce villages in the middle and town planners they have different characteristics we allow people to self- select and then we create what's known as a system of theft so the town plan is still from The Villages which forces the villages to Ste the work of the explorers and the Explorers therefore have to get on with exploring and this enforces Evolution into a system so I introduced that in all 2006 2007 there's a wonderful paper by uh gchq which is our intelligence Services which uh called boiling frogs you'll see a lot of those Concepts uh within that paper as well I'm cited in it so um yes but um I can talk about organization but I I wanted to talk about the sort of value and um the the um how do you use U maps to ask questions so let me just jump to something else hang on a second um Miro all right so this is a group uh they built a system

and like many groups they built a system the system had gone live and then somebody said you asked the question what's the return of investment and of course um at some point in the past somebody created some sort of plan which had a return investment calculation but if you dig into it what really happened is an executive read an article in some magazine or talk to a friend thought this was a good idea we should do some AI or whatever um had commissioned a group internally who then gone and talked to another group to find out how much would be the cost of implementing this then then taking that figure to another group to work out what the return investment calculation needed to look like and finally gone to marketing and say look can we have some evidence to justify building this and so the entire exercise was an exercise um in justifying building what the executive wanted on a whim and that's probably normal as in the vast majority of things go on terrifying because there is no process of Challenge and So eventually somebody said you know we spent all this money on this thing what is the value and and there is obviously some business case which gives a blahy blah and of course it's meaningless um so they asked me well how do we work out the value of the system so I simply said and this took two hour in total um first of all tell me what you think the system provides what do we mean by you know what is the value and so creating opportunities to add value quickly being able to provide synchronized environments that's apparently what this system was doing being user Centric okay cool out of the box solution for developers management of complicate all right fine so you got a B whole bunch of words and then what you do is I normally get them to group those words into themes or perspective perspectives so some of these words were about speed uh some of them were about cost and value some of them about removal of heavyweight Assurance processes some about simplification some about confidence in the environment now I often do these with research groups of you know sometimes fairly small 20 to 30 people more recently there's usually 80 or 90 people so we'll get huge amounts of words and and also these themes as well and then would choose a number of themes if a small group we might just pick one um if we have a large enough group we can break into smaller groups to to look at many so in this case they pick the removal of heavyweight Assurance processes which was not something they' been thinking of beforehand but something which came up in this discussion so we immediately went to mapping out from the perspective of our common platform for Assurance um and dated 2023 uh so a couple of things maps have no concept of time on them any more than geographical maps have what happens is that over time the map changes okay secondly or maps are imperfect representations of a space uh a perfect map of uh France would be or Paris

even would be the size of Paris which means as a map that's useless so all maps are imperfect representations in order to be useful you know but they are representations of the landscape they are not the landscape um the third thing is that all maps have a perspective so if no one had map Paris and you sent a group of people to map Paris and they came back and you ask them a question like what's the most important thing in Paris they might go Pierre's Pizza Parlor because they mapped it from the perspective of nice places to eat pizza so if you want to map space you send normally multiple groups to map from different perspectives and then what you can do is aggregate those perspectives together so you can you basically ask the same question on each map and then you aggregate the answers together so in this case they built a a map with a common Assurance uh platform they talked about what's involved engineering teams end users and once they hav't that we simply ask the question you know value assumes we going to be able to measure something what is it that we're going to be able to measure with this map and so they came up with a whole bunch of things like bugs between environments cost per transaction time for a new service uh decrease in negative press stories speed of assurance approval and once you've got those answers um you simply say how easy are they to implement and what's the value to the sponsor and users and you they just put them on here of course done in Mirror this entire process takes about 2 hours and now they've got a whole bunch of you know fairly easy to implement high value things that they can actually go and build metrics for and they had no idea beforehand what the cost per transaction or percent of uh policy implemented or time to fix for bug issues these are really simple things which they can now go and fix let's show you a slightly more uh demanding one so let me have a look agriculture let's look at sustainability sustainability okay yes um are you into sustainability or health care or what we do sustainability so this is a slightly larger one and this takes a long time this took hours um which you know in business terms is you not huge Sly enough people spend mons writing strategies which is quite funny so they started off by you know what matters in sustainability and you get this Splurge of words um things like modularity energy security protecting biod diversity great so this is the list here apparently it all matters um so you got to do all of this stuff green skills middle impact on resources loads of things so the next thing you do is you get people to group it into thieves uh so the first bit takes about an hour this bit takes about an hour so they created a whole bunch of themes or perspectives so policy standards and legislation some were connected to Circular economy some connected to leadership in governance organizational governments social all sorts of things and then what they do is you self organize into small groups and each of those groups

now goes and Maps out the industry from a particular perspective so this one is doing uh sustainability uh oh that was just an example sorry this one is doing modeling and complexity there's different people in the group and Tiani and Ben have taken the role as lead mappers and modeling and understanding complexity um within the sustainability field so they start adding components and then they usually start shifting to a tool like online worldly maps and so they're going through having discussion about what components matter where they matter Etc until they end up with what they think is a good enough map and the next question you ask is well if you've got a good enough map which is all you can ever have um where could you invest on on the map not where should we invest but where could you invest so they come up with a list of different areas that we could invest on the map and you do that each of these Maps um you can see they're picking different areas great then what you do is you aggregate this all together so you ask them uh a question um so um once they pick the different areas you ask them to highlight in their groups what are the most important areas for um and it could be improving Society or um return on investment making money and so they the Highlight in this case is for improving Society they've got different ideas so you've got one group collaborative Finance pattering characterization one's all about measurement and and surprisingly a supply chain audible tool chains and what you find is on the different Maps similar themed areas have come up so now you can aggregate across all the different maps and what you end up with something that looks like this so when we ask people and this it took 10 hours to get to starting from uh you know what do we do need to do about sustainability um it turns out and these are all people who work in the sustainability industry and there are about 60 of them from all sorts of different uh Parts turn out the things that we really need to focus on are Civic engagement compliance and auditing measurement and then after that manufacturing practices government policies Financial now interestingly when you ask the question or you go and do analyst reports so you aggregate analyst reports you tend to get the answers on the right hand side so the high priority is Technology Innovation we're always in the sustainability field investing in looking for we've been doing it for 40 years looking for the magic technology that will solve sustainability problems at the same time of course CO2 has gone up through the roof or it's going to be financial instruments like uh um carbon offsets again we've been doing this for 40 years getting nowhere um and again the marketing messages or the markets message is the stuff on the right and but you see a real difference once you once you map a landscape and ask people to focus on from the purpose of improving Society where would you focus it's very different from where would you focus in terms of

making money and you see that across so many different Industries Healthcare agriculture uh defense and all the rest of it um if you look at something like healthare oh there we are you end up with this so if you got a whole bunch of doctors and nurses together and you you get them to map out the space and everything else and um of course they have to break into small groups because you only have eight people are mapping a group at most usually so we try and break into small groups and then you aggregate them together what you discover is that if we want to improve health care uh from the focus on benefit to society it's all about measuring of Health outcomes auditing against prompts patient reported outcome measures which we don't do we measure against claros which is clinician report outcomes so symptoms we don't really have a health care System we have a sick care system we're very good at treating symptoms not making people healthy um so we should really focus on measurement Health outcome sharing a medal data and that's the list over there but if you look at it from the market it's all about personalized do it yourself medicine education well-being uh industrialized AI lot that's quite a high priority thing where of course you know and things like measurement of Health outcomes right at the bottom of the list so you often find those sorts of things so one of the things I do the maps is um it's not just when you want to look at a landscape if I can I will get a large group of people together well come up with some general themes by asking the question of what matters and then consolidate it and then we'll break into small groups like multiple sensors and we'll go and map out the space then we can ask each group what's important and then we take all those answers and we aggregate it together and this is what gives us a signal for um what matters so if you go back to the P example if you sent out a dozen groups to map from different perspectives rather than getting Pier's pizza parlor if you aggregated you're likely to find the Eiffel Tower for example point I will stop did that help oh and Susan you did the the education one which was always great because the mapping exercises themselves were always fun because the education group had a a bunch of people who worked in the education profession obviously and they always started from a focus of um uh whenever we were mapping it was benefit to students and opportunity and every time we map we were forced to basically recreate the map because there was the realization through the process the entire education system system doesn't work for the benefit of students it works for the benefit of government and social cohesion and the educational institutes and I mean it was the same with the the the sort of um Healthcare everybody thought wow Healthcare obviously is there for the benefit of the patients if you look at most the maps it's actually there for the benefit of medical institutions and pharmaceutical companies and

patients are right down the bottom of the list so yes we could expand that one I I will I was also on the gaming one oh yes what was interesting was that I think you mentioned that that was the only one that there was a symmetry between what the um what was good for culture and what was good for industry it was a wise investment as well as that which yeah I want to come back to because I think this kind of pulls this back into the the free energy if I look at the map for the gaming one so um Daniel I don't know if you've popped up the map but just to go through it we start off with this huge amount of noise of stuff that matters we going go through this process of categorizing into themes and generating the maps breaking into small groups mapping out the areas identifying on each of the maps what matters and not actually showing you that I can show you it's the stuff above oh here we are that's the all the groups doing their work identifying different themes and everything else and it's even more messy than that there we are um but basically what we end up with if I can find where I was there we are we end up with all this noise and we end up with this little table at the end created all together and it turned out yes you're absolutely right they quite a light I mean and probably the only example I've ever found uh where if you focus on what's important to society and if you focus what's on important in making a return on Capital there's pretty good alignment uh between the two I mean obviously AI content quite high in terms of making return on Capital but not bad Community is quite important for both that's slightly more important is the highest priority for uh a benefit to society um uh but uh it's not bad alignment at all uh certainly health care defense agriculture are way different really distorted well and I do want to point out that all of this is in GitHub it's all open source yes results of the maps um and all free yeah and you and so that you that uh speaks to what you know what I'm gonna point out what I'm what's top of mine and then I want to hear what what Daniel has to say but yeah specifically going back to what you said about um the the value versus the ease of implementation and you know and so when you're aggregating you collective intelligence then that's actually giving you predictive power in terms of where is the likelihood things are going to move um you and and and so the tapping into that um the self-organization the power in self-organization to enable that emergence of what is what is going to be the the the most powerful least uh you variational energy that we're going to be able to mobilize so it's mobilization um you know whether you're you you're in the midst of it now you know with active inference we're you know we're talking about an ecosystem with very very diverse um you know different professions and and so you they organized chaos but yeah I'm kind of seeing some routes in how to actually orchestrate those uh to find that you that sweet spot what do you thinking I I

think there there is another form of mapping which is created by a friend of mine Dave Snowden so Dave Snowden we meet up regularly like every week uh and uh we were talking a few hours ago actually uh before this this event so um and of course Dave's taken a sort of long interest in sort of mapping stuff that I've done he's got a a different way of looking at it it's um something called esterne mapping and that really looks at um because his focus is on you know if we've got a whole range of things we can do what are the things which are within the realm of what we can possibly do what what what in terms of energy input in terms of time uh becomes uh possible and so he's got some really interesting stuff about history mapping I mean there are there are very few forms of mapping that I'm aware of there's another one called uh Burger mapping which is really interesting if you're looking at stakeholders and political play in organization there's EST mapping as well and then there's my former wle Maps which um you know this this stuff is almost 20 years old now ridiculous but uh and this stuff by the way is all creative comments so it's interesting because once you create a map of a space and you've ID Ed the map to identify where you could invest and then you aggregate um to see in this case we're aggregating them in terms of what should we what is a high priority this isn't actually answering the question uh can we actually do this so um if you go back to the if I go back to the value one um so that was not education gaming let's have a look um no I don't want to open that one just common develop here you've got um where actually having identified different areas we're now asking that question of ease to implement versus um uh value to the sponsor so this is the point where you know you use the map to identify different areas and um you're now asking you know these are things we could easily change this helped you make that decision on what you should change I think EST mapping is a better form of doing that a bit here I mean I would use the maps to try and understand the landscape as much as possible and use the maps to sort of my form maps to identify the areas you could change help you develop the questions U but then in days form uh which which is EST mapping which is not this um really asked that question about the energy and the time I think that's a really interesting way of looking at the problem yeah and and uh yeah compare that to the typical cost benefit analysis oh God no which it's like you may as well use a SWAT diagram um okay I remember I've been doing this for 20 years so I mean I see well it's almost 20 years I the insurance example is typical example that I will see a group of people who've spent six months working on a problem with no actual understanding of the landscape and and they've come up with a plan to invest Millions into Robotics and uh they've got all the business case all the calculations all the vendors they've had in they've done all their due diligence the whole

thing's ready to go um and Beyond once we got over the 20 minutes of arguing it took 15 minutes of mapping and about five or 10 minutes of discussing the map to realize the entire thing was a waste of time that is incredibly common I um you know look before you leap I am always stunned um it's a constant source of marry of amusement um how many people make decision ions without um understanding the landscape and I do get myself into hot water so I I better warn your listeners um mapping will tend to get you fired the the reason being is that whilst I say it is possible you know to to once you understand a landscape and you can challenge and ask questions and all the rest of it um you do get you know you do get these moments where you you map out the environment and you realize the entire strategy of the company is to walk the people off the edge of a cliff and you want to point out that's really bad but it is really it's something the executive have been totally sold on uh and so they people get you know despite the fact they can see it clearly in front of them now it's almost they go into a sense of denial and and the solution is obviously you're a troublemaker so I've seen loads and loads of people uh well well not loads quite a few people get fired as a result of of mapping pointing out the insanity of something that is actually actually going on so I have to give you a a big healthy warning uh there but on a personal note I I have seen I've seen so many projects and I have got myself in trouble um I had one particular project uh they' asked me over to the US and I was sitting in this boardroom and uh this was about 10 and they taken me through their \$1.2 billion plan to build a private cloud and I'd map this out during this meeting at the end I'd turn around and uh I said I can build the entire lot for you with the same effect the 25 million and they were like how is that possible How would you do that and I say it's very simple you pay me 25 million I will sit on a beach for 5 years drinking margaritas then you phone me up and I will say we failed and that way you'll save all the rest of the money now this went down like a balloon I immediately got kicked out of the meeting all that sort of stuff it's spectacular um but what I what I most remember about this story is is about seven years later sex phoned me up and basically said we wish we paid you 25 million um of course I've wished they'd pay me 25 million as well but um this stuff is frighteningly common I'm afraid um I think that's actually where we can help because what you just said is what we're modeling it's called belief updating and so just look instead of instead of um focusing on behaviors yeah we're we're actually looking at the justifications um and it's quite revealing so well that that that that that's entirely different threat and um and leads into I should say an entirely different thread um which is the whole sort of because you can map culture as well I mean I I get into to um so I keep a lot of my maps on GitHub uh they're easy to find there's a

wly mapping or repulser there's a tool called online W bats you can just take the code and throw it in there again this is all Creative Commons open some stuff you just help yourself um but I also um do a lot of mapping of uh different environments so if you think about uh the concept of Defense so mapping out the defense space um there are multiple Landscapes uh on which sovereignty has a meaning uh the one that most people know about is territorial sovereignty So This Is Our Land um but there's also economic sovereignty uh technological sovereignty uh social sovereignty and political sovereignty and they're all important Landscapes which you can map now most countries I say most not all are the same should we say um do a reasonably good job of territorial uh and and pretty poor job of economic actually understanding their economic landscape very few actually understand their supply chains uh there's a wonderful piece of work on in Hungary which was looking at um a used vat records sales tax records to to basically graph out not even map but graph out the uh Hungarian economic system and of course they discover that around with these check points with these companies within there it's pretty terrifying it was a very small number of companies which if they failed would take out 25% of the GDP of the entire country just like terrifying um um but but of course most people are oblivious to this and of course technological I mean we're now finally talking about spawns but these are just graphs I say that's software building materials but they're not even maps of the space so we still don't maps of our technological landscape and political and cultural I mean most people of nowhere I mean um and and political I should say social and of course that's matters from the point of view of um sovereignty I mean uh you know if we cut back on foreign office Aid and all this sort of stuff we have to buy more bullets the two the two are not um independent they are correlated what you do in one landscape can affect other landscape so if you you decline you reduce soft power um you might have to exert more kinetic force in a in a territorial um but this this whole sort of area led me to um discussions uh where it was looking at how to map culture and that was fascinating um mostly because when I looked into culture I was thinking culture I mean what you know it's a really simple concept foolish foolish man that I was H and I I found um I think it was kobber he basically said anthropologists who are the experts on culture being arguing over what culture means uh for over 150 years or something it was a ridiculous length of time uh and of course there was no actual agreement on what culture meant which was for me quite shocking and um yeah this one do and then I came across a really really wonderful Anthropologist Margaret me really incredible stuff and um one of the statements of Margaret me was that um language was part of culture and that's interesting because if language is part of culture um and we we

therefore probably can't describe culture in terms of language uh that's something called Girdles in completeness theorem so no no model can be true and complete within itself so language is part of it we've got to find a different way so what I started doing was well can we map out cultural systems um so when I use a map I better just show you a couple of bits the the AIS at the bottom I always say Genesis custom product commodity uh when I'm looking at activities but this really you should say stage one two and three and four but you know that doesn't mean anything to anybody and this piece of work came from a a whole bunch of um text analysis of 9,730 no 9,231 documents back in 2005 2006 very manual process back then and it turns out there's these different stages and they have different labels um different characteristics as well so the different labels are when we talk about practices novel emerging good best data model Divergent convergent model knowledge concept hypothesis Theory and then something comes accept and I can pick any label from stage one and mix it with stage two and three and four so I started mapping out things like ethical values so looking at you know was it a concept was it an emerging concept was it sort of converging was it accepted so he's doing a lot of stuff on um sexual identity and mapping out in that space as well uh so when you look at turns out you start looking at a collective and you will'll have various beliefs in a collective it might be Universal basic income paid holiday unionization uh and those beliefs themselves will have different stages of evolution and they'll be based upon um underlying concept workers rights anti-discrimination civil rights both of those extend from the abolition of slavery and underneath that are Concepts like reciprocity and fairness um so you you can see how these you know these components themselves have evolved we have a fairly good idea or used to I should say a fairly good idea of Concepts like reciprocity um that's being a little bit challenged at the moment but if you take this this map and you just say well Okay so we've got some values beliefs and we start off with this point here can you I don't know if you can see my mouth um it's values in the middle okay so what I did was I I had a blank map with Concept em merging converging accepted at the bottom and charted industrialize and put values as you know there's this map of values in the middle put that as a square box and then I got a group of people who couldn't agree on anything uh these were brexiteers and remainers in the UK brexiteers were like leave Europe remainers wanted to remain and these group were incredibly hostile to each other so if the brexiteers said we'll have a cup of tea the remainers would go will'll having coffee and if the Romanians won't well have Biscuit the brexiteers would go nowhere not having biscuits for cake I mean these people would just not agree on anything whatsoever um so I had them in the room and I said well

look I'm trying to explore the concept of culture and I know I can't do it through the use of a word or words alone so I've got this concept of values and that's a that's a map itself and I've put that in the middle I want to think about what are the components that make up culture and they were a bble to discuss this rather complicated issue and Le through the meeting the map and this is where they end up with so at the very top the anchors or we and me we're all a bit we we're all a bit me as the individual and Collective uh the we is all about control of the environment me is all about individuality which is about agency which is connected to Concepts like power and control of course is uh you know connected to the collectives that we belong to we belong to many forms of collectives and all collectives want to succeed in the spread of their values a church a football club a family or whatever you know they all want to succeed um and of course our behaviors are a combination of the values that we have and the doctrine or the principles we apply so for example um a principle is focus on user needs well that sounds good but but if your belief is that people with blue eyes and blonde hair are somehow better than everybody else then the behavior are focusing on user needs and the belief that blonde blue-eyed people are somehow Superior is not a very appealing behavior um but um then you get into concepts of safety which depends upon behavior and sense of belonging which depends upon behavior and so you start getting Cycles within this and you can actually use this to to form groups also destabilize groups as well because you that's one of the things you do with can do with maps as well but that's how I many many moons ago gosh this must be about eight years ago um I got into mapping or maybe less seven years ago uh mapping into cultural systems and this turns out to be a a a very imperfect but interesting definition of culture so when I say when people say what is culture I go well that's culture and um my particular interest and the reason why we did the video gaming one is because if you want to change the behaviors of a collective well the principles are are universally useful principles so as long as people are aware of and there's a list of about 40 of those what we need to really do is change the values and the beliefs and one of the ways of doing that is through memory so symbols rituals and heroes um or in other words art art is one of the most powerful forces there is to changing the values and beliefs in are collected and the single largest form of art today is immersive video games which is why we were doing that research project because I have a particular interest if we want to change the behaviors of you know National collectives um probably the best way of attacking the problem is through andless of video games that all makes sense yeah thank you I mean there there's so many places to jump in but I want to respect your time perhaps we can close the session there and then Susan and I

could stay for a few minutes and read some of the thoughts that we had about active inference
oh my apologies I didn't see what the time was oh sorry I talked for far too long I I have a
terrible habit I can do this for days um I hope I hav B your rigid um you said what you wanted
to say oh no was fantastic fantastic and I feel would you like to continue a bit longer or I can
just mention a few of the points that Susan and I had like thought about and prepared just more
to footnote it for just thinking about it more in everyone's own time um yeah there there's so
much here that uh that you there is so many different crossovers between uh these these two
directions yeah and uh and and so you that I mean I've got so many but I'm GNA let turn it
over to uh to Daniel to kind of pull these thread it all together so we can um mine this uh the
Brilliance this is just going to be reading from the document that we already prepared so not
just to connect and and notes being taken so it's not in any order this isn't any specific
synthesis um but the the the ideas of cognitive mapping worldly mapping as a cognitive
activity strategy or behavior being separated from the landscape this is like the agent on the
niche the phone distribution and the nestmate and the pherone distribution can't be changed
instantly that's actually something happening that's separate from the control of like any
navigation path and landscape and so being able to have real specific practices and all of this
fractal diversity and practices as well just like there's Tools in a workshop in the path finding
wayf finding areas this is mapping technique and then underneath cardgraphic technique is
cardgraphic research and measurement and Signal processing control theory all these other
areas um however more specification or formalization is not always the rate limiting step or
the step that's aligned with any particular outcome that's kind of like the recognition that
actually I think just from what I have heard helps the participation go beyond just sourcing
information and words like if you just wanted all the words you could download a dictionary
but there's actually something in the process and the the time and the attention as that feeds
back upon itself that is is qualitatively different than just like the Playbook the playbook in the
kitchen you know is different from the restaurant changing through time so like that those are
just kind of General then just a little bit more on the specifics with active inference really
interesting how this is like a snapshot shot that has position and then a first derivative or
second derivative like how how fast is it changing what Vector is it changing in how fast is
that changing that could be taken very far formally with active inference that's basically basian
mechanics then separation of consideration with the UDA Loop and the connection to the
principles also previously we've connected the UDA functions and cycle to action perception
basically describing the same territory they're different Maps so having a PL ISM of

approaches and actually mapping them at a higher level like systems engineering Maps or active inference ecosystems of shared intelligence complex systems all these being recognized as Maps helps have the epistemic humility and the pluralism and it's also like arguably a correct formalism that takes mapping to like its logical completeness as a method of pragmatic inquiry and then um separation of of the sense making of the landscape from the actions that can be taken and then in active inference like the equations have all those kinds of considerations like how the world States change through time that the perceptions the actions the affordances like all these different terms um however that equation is very um silent on how to do it procedurally through time with a group obviously the equation could be used in in every single possible way so even if like this were more accurate that you could make a decision with this or that consequence on this map or you could have chosen the other way on a different one and so then that brings in some really interesting considerations of inquiry which I think just to kind of conclude speaks to the different ways that some of the theory of cognitive systems and all of like the connections that people will see who study the formalisms in act dim that theory and how that remarkable seeing what people you know think and how they get involved so thank you bye Awesome by bye